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Numerical Results Report of the Second Assignment

Aeronautical Engineering - Structural dynamics and aeroelasticity course

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Introduction

This report presents the numerical results for the aeroelastic and dynamic analysis of a strut-braced wing utilizing a Ritz-Galerkin discretization scheme implemented in `MATLAB (R2025b)`.

1. Convergence Analysis

The continuous wing layout is modeled using a Ritz-Galerkin approach, where the transverse displacement $w(y,t)$ and the torsional twist $\theta(y,t)$ are approximated via a truncated set of shape functions: $\phi_j(y) = (y/b)^{j+1}$ $(j = 1, \ldots, N_w)$ and $\psi_j(y) = (y/b)^j$ $(j = 1, \ldots, N_t)$.
To determine the minimum number of modes, a convergence study was conducted on the first two structural natural frequencies (ω_1: bending; ω_2: torsion) with a relative error tolerance of 10^{-3}. Table 1 details the convergence history.

Table 1: Frequency Convergence.

N	N_w	N_t	ω_1 [rad/s]	ω_2 [rad/s]	ϵ_1 [-]	ϵ_2 [-]
1	2	1	20.742801	53.341752	—	—
2	3	2	17.774097	40.771403	1.67e-01	3.08e-01
3	4	3	17.764031	40.129922	5.67e-04	1.60e-02
4	5	4	17.589844	39.813284	9.90e-03	7.95e-03
5	6	5	17.581427	39.445767	4.79e-04	9.32e-03
6	7	6	17.543611	39.286711	2.16e-03	4.05e-03
7	8	7	17.540277	39.139559	1.90e-04	3.76e-03
8	9	8	17.528776	38.911658	6.56e-04	5.86e-03
9	**10**	**9**	**17.528243**	**38.887630**	**3.04e-05**	**6.18e-04**

As illustrated by the error trends in Figure 1 on Page 3, the relative error metrics drop below the target threshold at $N = 9$ $(N_w = 10, N_t = 9)$, establishing a converged basis with $\omega_1 = 17.5282$ rad/s and $\omega_2 = 38.8876$ rad/s.

2. Frequency Response Function (FRF)

Using the converged structural matrices at $U_\infty = 25$ m/s, the Frequency Response Function (FRF) of a harmonic aileron deflection input $\beta(t) = \beta_0 e^{j\Omega t}$ ($\beta_0 = 1$ rad) to the total physical displacement at the wing tip ($y = b, x_P = -0.5c$) was evaluated across a frequency sweep $\Omega \in [0.1, 1000]$ rad/s.
From Figure 2 (Page 3), the magnitude spectrum outlines two distinct resonance peaks. The first amplification peak occurs near $\Omega \approx 17.5$ rad/s, coinciding with the first fundamental bending mode. The second resonance peak stands near $\Omega \approx 38.9$ rad/s, matching the fundamental torsional mode. The phase response shows the $180°$ drop across each resonant frequency.

3. Aeroelastic Stability

Aeroelastic stability boundaries were explored by examining the critical airspeeds for flutter (V_{flutter}) and structural divergence ($V_{\text{divergence}}$) as a function of the chordwise attachment position of the strut x_B, normalized by the chord length c ($x_B/c \in [-0.5, 0.5]$).
As shown in Figure 3 (Page 3), the baseline configuration ($x_B/c = -0.5$, Leading Edge) provides a high flutter threshold of $V_{\text{flutter}} = 65.90$ m/s. Shifting the attachment point aft drastically changes the coupling: the minimum flutter speed drops to $V_{\text{flutter, min}} = 26.56$ m/s at $x_B/c = 0.096$. Past $x_B/c \approx 0.17$, the static divergence speed $V_{\text{divergence}}$ drops below the flutter curve, becoming the dominant instability mechanism.

4. Gust Response

Gust response analysis was performed at $U_\infty = 0.75 \times V_{\text{flutter}} = 49.425$ m/s, using direct state recovery to evaluate the root bending moment $M_x(0, t)$.

4.1. Deterministic Unit Step Gust Response

First of all, the system was excited with a unit step gust ($v_g = 1$ m/s).
From Figure 4, subjected to the sudden increase in lift from the step gust, the root bending moment exhibits a transient overshoot, peaking at around $M_x \approx 4700$ Nm within the first 0.2 seconds. Due to the aerodynamic damping, these transient dynamic oscillations decay very rapidly, completely settling down within approximately 1.5 seconds to the steady-state asymptotic static value of $M_{x,\text{static}} = 1684.7$ Nm.

4.2. White Noise Gust Analysis

To model continuous atmospheric turbulence, the wing was subjected to a continuous unitary white noise gust field. The statistical properties of the root bending moment were calculated via direct state-space covariance and cross-verified via the analytical Lyapunov equation:

$$A_{\text{ly}}\sigma^2_{\text{xx}} + \sigma^2_{\text{xx}}A^T_{\text{ly}} + B_{\text{ly}}WB^T_{\text{ly}} = 0 \tag{1}$$

The results are summarized below:

- Root Bending Moment Variance ($\sigma^2_{M_x}$): 3.9460×10^9 Nm^2
- Root Bending Moment Std. Dev. (σ_{M_x}): 6.2817×10^4 Nm
- Lyapunov Validation Variance (σ^2_{lyap}): 3.9460×10^9 Nm^2

The mathematical matching between the state covariance and the explicit Lyapunov solution confirms the numerical consistency of the state-space realization.
From a physical standpoint, the resulting variance and standard deviation values are remarkably high. This outcome is a direct consequence of the ideal white noise approximation used for the gust input. By definition, a pure white noise process possesses a constant power spectral density across an infinite frequency spectrum ($f \in (-\infty, \infty)$). Consequently, it delivers energy to and excites all structural modes and high-frequency dynamics of the system simultaneously.
In a real atmospheric environment, this condition is physically impossible, as real gust spectra (such as the Von Kármán or Dryden models) exhibit a roll-off at higher frequencies, meaning that the turbulence energy decays and does not excite the high-frequency spectrum.

Report Diagrams and Plots

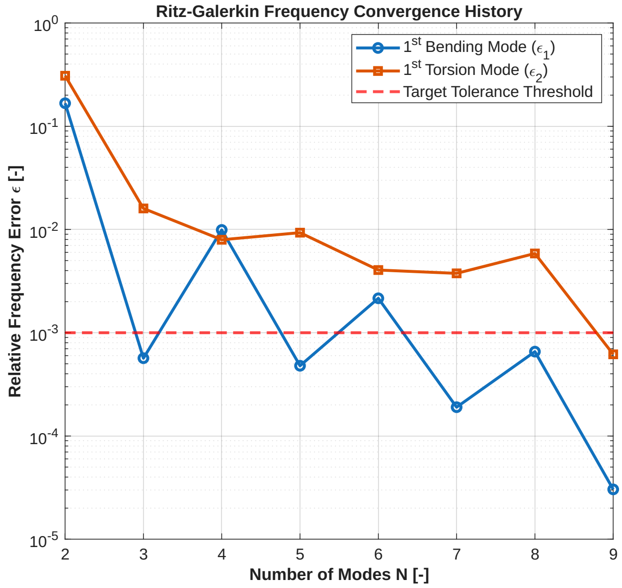

Figure 1: Ritz-Galerkin frequency convergence history (ϵ_1, ϵ_2).

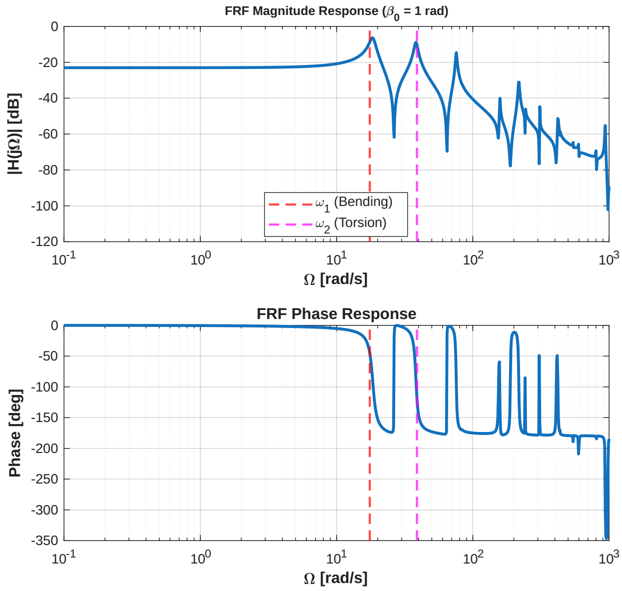

Figure 2: Bode diagram of the wingtip displacement FRF.

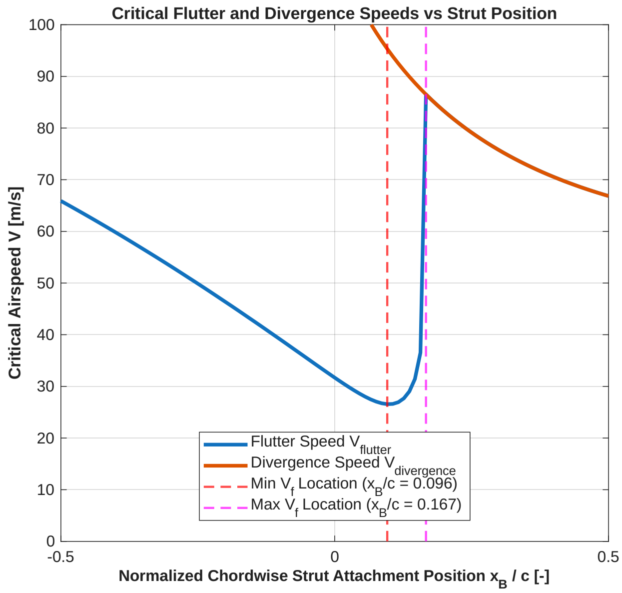

Figure 3: Critical airspeeds vs normalized strut position x_B/c.

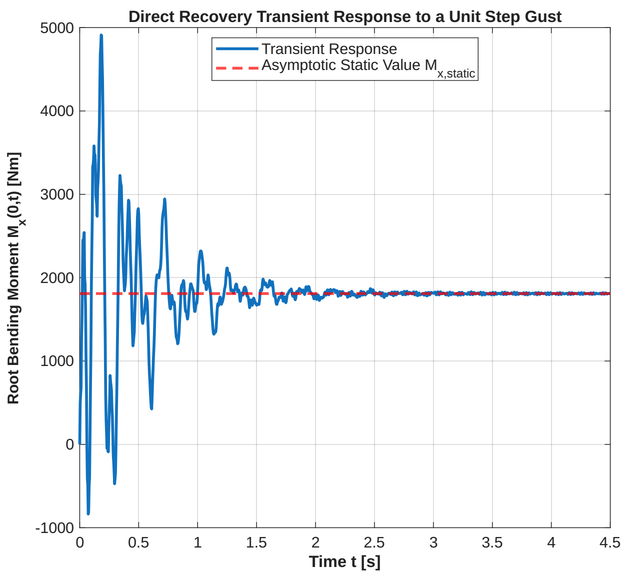

Figure 4: Root bending moment $M_x(0, t)$ transient response.